

Source-Correlation Image and Finite Signed Feasibility in Literal Twin-Prime Shells

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Abstract

TPC-294 found strong cancellation in an ambient equal-sign cube for the exactly weighted Gram form, but did not determine whether its favorable sign targets could be produced by a source-side correlation map. This paper isolates that finite linear-algebra question. Let the physical shell vectors be the columns of a rational matrix A and let $C = A^\top$ map source coordinates to prime correlations. We prove that nonsingularity of $G = A^\top A$ makes C surjective and gives the explicit witness $h = AG^{-1}b$ for every target b . Two independently replayed modular determinant certificates establish full column rank on all 18 inherited rows, covering 1,380 shell edges. Consequently every TPC-294 weighted minimizer, unit-edge max-cut target, and all-positive target has an exact witness in the declared unrestricted finite rational source space. This closes only the broad finite image question: admissibility in the frozen Mobius/comparison profile, witness-norm growth, arithmetic L^2 , and the twin-prime endpoint remain open.

1 Position on the route

The project studies one literal prime-shell dynamical family throughout. Earlier releases separated physical prime components and measured their cross-prime Gram interaction [5, 3, 2]. Adaptive nonnegative weighting encountered a coherence wall [1]; signed pair cancellation was then tested [7]. Three-prime parity and the complete signed-shell max-cut layer exposed combinatorial obstructions [8, 6]. Restoring the true Gram magnitudes in TPC-294 showed that the weighted sign optimizer can be substantially better than the unit-edge surrogate [4].

That last result left a precise question. The signs in TPC-294 were chosen in an ambient coefficient cube. Are they even in the image of a source perturbation? The present paper answers the finite question for an explicitly declared broad source space. The answer is positive because the inherited physical columns have full Gram rank. The qualification “broad” is part of the theorem: no assertion is made here about the frozen arithmetic source profile.

2 Finite physical shell and source map

Let $I = I_N$ be the frozen integer interval. For an odd prime q , height H , and exponent s , write

$$K_{H,s}(v) = \frac{H^{2s}}{(H^2 + v^2)^s}, \quad B_q(u, t) = \mathbf{1}_{q \nmid u} \mathbf{1}_{q \nmid t} \left(\mathbf{1}_{u \equiv t \pmod{q}} - \frac{1}{q-1} \right).$$

The deleted-diagonal physical component inherited from the frozen source vector is

$$(g_q)_u = \sum_{t \in I, t \neq u} q K_{H,s}(u-t) B_q(u, t) \beta_t, \quad u \in I.$$

For a finite shell $S = \{q : Q < q \leq 2Q\}$, set

$$A = [g_q]_{q \in S} \in \mathbb{Q}^{I \times S}, \quad G = A^\top A.$$

Thus G is exactly the Gram matrix used by TPC-294. The source-side correlation map is

$$C = A^\top : \mathbb{Q}^I \longrightarrow \mathbb{Q}^S, \quad C(h) = (\langle h, g_q \rangle)_{q \in S}.$$

The source space in this release is the full finite rational coordinate space $\mathbb{Q}^I$. Restricting it to a parameterized arithmetic profile is a different problem and is not silently built into C .

3 Exact image theorem

The basic observation is elementary but decisive for the route map.

Theorem 1 (Full Gram rank gives a full finite source image). *Let $A \in \mathbb{Q}^{n \times m}$ and $G = A^\top A$. If G is nonsingular, then $C = A^\top : \mathbb{Q}^n \rightarrow \mathbb{Q}^m$ is surjective. For every $b \in \mathbb{Q}^m$, one exact preimage is*

$$h_b = AG^{-1}b.$$

Proof. Nonsingularity gives G^{-1} over $\mathbb{Q}$. Substitution yields

$$A^\top h_b = A^\top AG^{-1}b = GG^{-1}b = b.$$

Hence every target has a preimage. □

Proposition 1 (The displayed witness is the least-norm witness). *Under the hypotheses of the theorem, the same vector h_b is the unique minimum- ℓ^2 solution of $A^\top h = b$ over $\mathbb{R}^n$.*

Proof. The vector h_b belongs to $\text{col}(A)$, while every difference of two solutions belongs to $\ker(A^\top)$. These two subspaces are orthogonal. Therefore every real solution has the form $h_b + v$ with $v \perp h_b$, and $\|h_b + v\|^2 = \|h_b\|^2 + \|v\|^2$. □

The proposition identifies the next quantitative obstruction: exact attainability does not imply that the least source cost is small enough for an analytic argument. In particular, the theorem pays no asymptotic power credit.

4 Modular rank certificate

All entries in (2) are rational on the declared finite grid. For a prime modulus p avoiding their denominators, reduce G entrywise to G_p . We use the following denominator-clearing fact.

Lemma 1 (Nonzero modular determinant). *If every entry denominator is invertible modulo p and $\det(G_p) \neq 0$ in $\mathbb{F}_p$, then $\det(G) \neq 0$ over $\mathbb{Q}$.*

Proof. Choose a common denominator D with $p \nmid D$. The integer $D^m \det(G)$ reduces to $\det(G_p)$. A nonzero reduction cannot come from the zero integer, so $\det(G) \neq 0$. □

The producer accumulates each g_q target-first, forms the rational Gram matrix, and performs Gaussian elimination modulo

$$p_1 = 1,000,000,007, \quad p_2 = 998,244,353.$$

The independent replay accumulates the same columns source-first and does not import the producer. Target residuals for the all-positive label, the TPC-293 max-cut label, and the TPC-294 minimum label are also checked modulo both primes. These residual checks are a replay guard; exact feasibility itself follows from Theorem 1 once the determinant certificate is accepted.

5 Inherited finite atlas

The audit contains eight $s = 2$ growth rows, four $s = 1$ crossover rows, and six $s = 2$ source-control rows. The shell cardinalities range from 3 to 17. The resulting finite image statement is summarized in Table 1.

Table 1: Finite source-image audit inherited from TPC-294.

quantity	count
rows	18
shell edges	1,380
full rank modulo p_1	18 / 18
full rank modulo p_2	18 / 18
unrestricted source image surjective	18 / 18
TPC-294 weighted minimizers feasible	18 / 18
all-positive targets feasible	18 / 18
unit-edge max-cut targets feasible	18 / 18
fixed-power credit	0

Table 2: Representative rows and inherited weighted minima.

axis	N	H	Q	s	$ S $	$\det_{p_1}$	$R_{\min}$
growth	128	24	9	2	3	346506124	0.5343
growth	192	32	16	2	5	316685990	0.4091
growth	384	50	40	2	10	242483411	0.1823
growth	512	58	60	2	13	277057017	0.1078
growth	512	58	90	2	17	98453757	0.04964
crossover	256	38	27	1	7	973455099	0.5191
control	384	48	70	2	15	109411839	0.05849
control	384	52	70	2	15	413185836	0.06383

Table 2 records representative rows. Determinant residues are shown only as compact identifiers of the exact modular certificates; all source targets were tested with zero residual modulo both moduli.

For example, at the late row $(N, H, Q, z, s) = (512, 58, 90, 5, 2)$, the shell has 17 primes and the two determinant residues are 98,453,757 and 233,740,651, respectively. At the exceptional crossover row $(256, 38, 27, 5, 1)$ they are 973,455,099 and 333,763,173. Both are nonzero, so the exact source-image theorem applies in each case.

The target transfer is not merely existential at the level of labels. For each row the certificate records zero modular residual for all three target families, and the exact witness formula is explicit:

$$h_{\min} = AG^{-1}a_{\min}, \quad h_{\text{cut}} = AG^{-1}a_{\text{cut}}, \quad h_+ = AG^{-1}\mathbf{1}.$$

Here $a_{\min}$ is the exact weighted minimizer from TPC-294 and a_{cut} is its unit-edge max-cut comparison. The statement is about unrestricted rational coordinates, not about the original frozen choice of β .

6 What this does and does not close

The finite atlas closes a genuine branch of the map: an ambient signed target cannot be rejected merely because the finite linear correlation map has too small an image. This is useful because TPC-294’s favorable weighted directions are now known to possess explicit finite source preimages.

It also sharpens the obstruction. There are at least three remaining filters between this theorem and a twin-prime result:

1. the witness must belong to the native Mobius/comparison profile or a rigorously justified enlargement of it;
2. the least-norm cost $\|AG^{-1}b\|$ must be bounded in a growing shell with the correct normalization;
3. the resulting source construction must feed the arithmetic L^2 estimate and the Route-B endpoint without spending more power than it gains.

The first issue is an image-of-a-profile problem, not a rank problem. The second is a conditioning problem: full rank can coexist with a very large inverse Gram matrix. The third is the original arithmetic bottleneck. Accordingly, the result is deliberately recorded as

$$\text{PROVED_EXACT_FINITE} \quad + \quad \text{NUMERICALLY_CERTIFIED_FINITE},$$

with no asymptotic exponent credit and no twin-prime conclusion.

7 Reproducibility and conclusion

From the project directory, run:

```
export PYTHONDONTWRITEBYTECODE=1
python -B code/tpc295_source_correlation_image_certificate.py --check
python -B experiments/tpc295_independent_checker.py
python -B experiments/tpc295_source_image_stress.py
```

The optimized versions use Python’s `-O` flag and must have byte-identical standard output and empty standard error. The canonical finite record is `results/tpc295_certificate.json`.

The Session-named Route-A/Route-B evaluator files are absent from this checkout. Therefore this paper reports the local proof package, exact canonical certificate, independent replay, stress test, and Bridge-B checker as the available fail-closed evidence; it does not declare an official evaluator pass.

The next route experiment is quantitative rather than another ambient sign search: evaluate the least-norm witness cost, compare it with admissible source-profile spans, and test whether any normalized budget can reach the arithmetic L^2 interface. This is the `ROUND2.CLUE` recorded by the certificate.

References

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